

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION with OpenCV
COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 10%

1. An image appears too bright. Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.

ANS: Image Appears Too Bright

Concept:

A bright image has high intensity values, making details difficult to see.

Suitable Grey Level Transformation:

Apply Negative Transformation (or Intensity Reduction) to reduce brightness and improve visibility.

Application Process:

The transformation changes each pixel using the formula:

$$s = 255 - r$$

This reduces the effect of excessive brightness.

Justification:

Negative transformation improves visibility by balancing the intensity values and making image details clearer.

Expected Outcome:

The image becomes less bright, with improved contrast and better visibility of objects.

2. A dark image lacks detail. Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning.

ANS: Dark Image Lacks Detail

Concept:

A dark image has low-intensity pixel values, making details difficult to see.

Suitable Grey Level Transformation:

Apply Log Transformation to enhance low-intensity regions.

Application Process:

The transformation increases the brightness of dark pixels using the formula:

$$s = c \log(1 + r)$$

Justification:

Log transformation brightens dark areas while preserving image details, making hidden features more visible.

Expected Outcome:

The image becomes brighter, low-intensity regions are enhanced, and overall visibility is improved.

3. An image has poor contrast. Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.

ANS: Image Has Poor Contrast

Concept:

A poor contrast image has very little difference between dark and bright regions, making details difficult to see.

Suitable Histogram-Based Technique:

Apply Histogram Equalization to enhance the image contrast.

Application Process:

Histogram equalization redistributes the pixel intensity values over the full intensity range, improving the overall contrast.

Justification:

It increases the visibility of objects and fine details by spreading the intensity values evenly.

Expected Outcome:

The image becomes clearer, with improved contrast and better visibility of features.

4. A low-contrast image requires uniform intensity distribution. Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.

ANS: Low-Contrast Image with Uniform Intensity Distribution

Concept:

A low-contrast image has pixel values concentrated in a small intensity range, making the image appear dull.

Suitable Method:

Apply Histogram Equalization to achieve a more uniform intensity distribution.

Working Process:

Histogram equalization redistributes the pixel intensity values across the entire intensity range, increasing the difference between dark and bright regions.

Justification:

It improves contrast and makes image details more visible by spreading the intensity levels uniformly.

Expected Outcome:

The image becomes clearer, with better contrast, uniform intensity distribution, and improved visibility of objects.

5. An image shows uneven brightness across regions. Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.

ANS: 5. Uneven Brightness Across Regions

Concept:

Uneven brightness occurs when different parts of an image have different illumination levels, making some regions too bright and others too dark.

Suitable Enhancement Technique:

Apply Adaptive Histogram Equalization (AHE/CLAHE) to correct non-uniform illumination.

Application Process:

The image is divided into small regions, and histogram equalization is applied to each region separately. This improves local contrast and balances brightness.

Justification:

Adaptive histogram equalization enhances both dark and bright regions without over-enhancing the entire image, making it suitable for uneven illumination.

Expected Outcome:

The image has more uniform brightness, improved contrast, and clearer visibility of details across all regions.

6. A noisy image contains random intensity variations. Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.

ANS: 6. Noisy Image with Random Intensity Variations

Concept:

A noisy image contains random intensity variations that reduce image quality and hide important details.

Suitable Smoothing Filter:

Apply a Mean (Averaging) Filter to reduce the noise.

Working Mechanism:

The mean filter replaces each pixel with the average value of its neighboring pixels, smoothing random intensity variations.

Justification:

It effectively reduces random noise and produces a smoother image, making objects easier to identify.

Expected Outcome:

The image becomes smoother, noise is reduced, and overall image quality is improved.

7. An Image requires noise reduction without significant loss of details. Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.

ANS: 7. Noise Reduction Without Losing Details

Concept:

The image contains noise, but important edges and details should be preserved during filtering.

Suitable Filtering Approach:

Apply a Median Filter.

Working Mechanism:

The median filter replaces each pixel with the median value of its neighboring pixels instead of the average, removing noise while preserving edges.

Justification:

Unlike the mean filter, the median filter removes noise effectively without causing significant blurring of image details.

Expected Outcome:

The image becomes cleaner, noise is reduced, and important edges and fine details are preserved.

8. An image appears blurred after smoothing. Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.

ANS: 8. Blurred Image After Smoothing

Concept:

Smoothing removes noise but can also blur edges and reduce image sharpness.

Suitable Sharpening Method:

Apply the Laplacian Filter for image sharpening.

Working Principle:

The Laplacian filter highlights regions with rapid intensity changes (edges), making the image sharper and restoring lost details.

Justification:

It enhances edges and fine features, improving image clarity after smoothing.

Expected Outcome:

The image becomes sharper, edges are restored, and important details are more clearly visible.

9. Fine details in an image are not clearly visible. Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.

ANS: 1. Fine Details Not Clearly Visible

Concept:

Fine details are hidden because high-frequency components are weak or suppressed.

Suitable Filtering Technique:

Apply a High-Pass Filter in the frequency domain.

Working Principle:

The high-pass filter removes low-frequency components and preserves high-frequency components such as edges and fine details.

Justification:

It enhances image sharpness and makes edges and small features more visible.

Expected Outcome:

The image becomes sharper, with clearer edges and improved visibility of fine details.

10. A noisy image must be preprocessed before further analysis. Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.

ANS: 10. Noisy Image Preprocessing

Concept:

A noisy image contains unwanted random variations that reduce image quality and affect further analysis.

Suitable Spatial Filtering Method:

Apply a Median Filter.

Working Process:

The median filter replaces each pixel with the median value of its neighboring pixels.

This removes noise while preserving image edges.

Justification:

The median filter is more effective than the mean filter for removing noise because it preserves important details and does not blur edges significantly.

Expected Outcome:

The image becomes cleaner, noise is reduced, and important details are preserved, making it suitable for further analysis.

Code of Conduct:

I AMALRAJ P (192524089) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rubrics for Evaluation

<i>Criteria</i>	<i>Weightage</i>	<i>Level 4 – Exemplary</i>	<i>Level 3 – Proficient</i>	<i>Level 2 – Developing</i>	<i>Level 1 – Beginning</i>
<i>Understanding of Concepts</i>	<i>25%</i>	<i>Demonstrates thorough, accurate understanding of concepts</i>	<i>Good understanding with minor gaps</i>	<i>Basic understanding with some misconceptions</i>	<i>Poor understanding; major misconceptions</i>
<i>Explanation</i>	<i>25%</i>	<i>Detailed, well-explained answers with relevant examples</i>	<i>Clear explanation with adequate details</i>	<i>Limited explanation; lacks depth</i>	<i>Poor or unclear explanation</i>
<i>Application</i>	<i>20%</i>	<i>Effectively applies concepts with strong analytical ability</i>	<i>Applies concepts with minor errors</i>	<i>Limited application with weak analysis</i>	<i>No application or incorrect analysis</i>

<i>Organization</i>	<i>15%</i>	<i>Well-structure d answers with logical flow and coherence</i>	<i>Organized with minor issues in flow</i>	<i>Basic structure, lacks clarity</i>	<i>Poor organization , incoherent answers</i>
<i>Accuracy</i>	<i>10%</i>	<i>Completely accurate and covers all required points</i>	<i>Mostly accurate with minor omissions</i>	<i>Partially correct with missing points</i>	<i>Incorrect answers with major gaps</i>
<i>Clarity</i>	<i>5%</i>	<i>Clear, neat, professional presentation</i>	<i>Understanda ble with minor issues</i>	<i>Limited clarity, readability issues</i>	<i>Poorly written, difficult to understand</i>